

Exploratory Data Analysis (EDA)

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Date: **March 15, 2026**

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a statistical approach used to examine and summarize datasets before applying formal statistical models. It allows researchers to understand the structure of the data, detect errors or outliers, and identify potential relationships between variables. The concept was introduced and popularized by Tukey (1977), who emphasized that researchers should explore and visualize data before conducting formal hypothesis testing.

EDA typically involves the use of descriptive statistics and graphical techniques. Descriptive measures such as the mean, median, mode, standard deviation, and range help summarize the distribution of data. Graphical tools—including histograms, box plots, bar charts, scatter plots, and line graphs—help reveal patterns, trends, and possible relationships between variables (Field, 2018; James et al., 2021). Through these techniques, researchers can gain preliminary insights into the characteristics of a dataset.

For example, when examining the relationship between students' study hours and exam scores, descriptive statistics can summarize the overall performance of students, while graphical methods such as scatter plots can reveal whether increased study time is associated with higher academic performance. These exploratory methods help researchers generate initial interpretations and guide further statistical analysis.

Generally, EDA is an essential step in the data analysis process because it allows researchers to better understand their data before applying more advanced statistical techniques. By identifying patterns, anomalies, and possible relationships, EDA helps

ensure that subsequent analyses are appropriate and meaningful (Tukey, 1977; Wickham & Grolemund, 2017).

Objectives of the Study

The objective of this paper is to explain the concept of Exploratory Data Analysis by describing four common types of analysis: **univariate analysis, bivariate analysis, multivariate analysis, and graphical (visual) analysis**. These approaches are used to examine datasets, identify patterns, detect anomalies, and explore relationships among variables.

Student	Study Hours	Exam Score
A	2	65
B	3	70
C	4	72
D	5	78
E	6	85
F	1	60
G	7	90
H	3	68
I	5	80
J	4	75

Table 1: Study Hours and Exam Scores of Students

Univariate analysis focuses on examining a single variable at a time. This type of analysis is used to describe the basic features of a dataset through measures such as the mean, median, and range (Field, 2018). In this example, analyzing only the **exam scores** allows researchers to understand the distribution of student performance and identify the highest, lowest, and average scores.

Student	Study Hours	Exam Score
A	2	65
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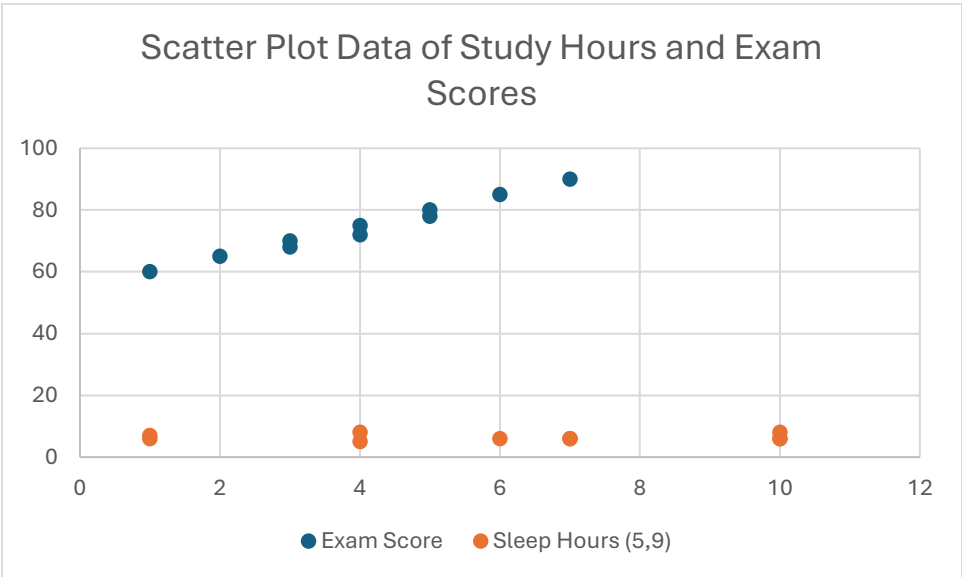
Table 2: Relationship Between Study Hours and Exam Scores

Bivariate analysis examines the relationship between two variables in order to determine whether they are associated (James et al., 2021). In this case, comparing **study hours and exam scores** suggests that students who spend more time studying tend to obtain higher scores. This type of analysis helps researchers identify potential correlations between variables.

Student	Study Hours	Exam Score	Attendance (out of 10)	Sleep Hours (5,9)
A	2	65	7	6
B	3	70	1	7
C	4	72	10	6
D	5	78	1	6
E	6	85	6	6
F	1	60	10	6
G	7	90	7	6
H	3	68	4	5
I	5	80	4	8
J	4	75	10	8

Table 3: Study Hours, Sleep Hours, Attendance, and Exam Scores

Multivariate analysis involves examining relationships among **more than two variables simultaneously**. This approach allows researchers to understand how multiple factors may influence an outcome variable (Hair et al., 2019). In this example, additional variables such as **sleep hours** and **class attendance** are considered along with study hours to explore how several factors may contribute to exam performance.



Lastly, **Graphical Analysis**—Graphical analysis involves the use of visual tools to better understand patterns and relationships in data. Visualizations such as scatter plots are particularly useful for identifying trends and correlations between variables (Cleveland, 1993).

In this example, a scatter plot of **study hours and exam scores** would likely show a **positive relationship** between the two variables. As study hours increase from approximately 1 to 7 hours, exam scores also increase from around 60 to 90. This upward trend suggests that greater study time is associated with improved academic performance. Meanwhile, sleep hours appear relatively consistent across students and do not show a strong pattern with exam scores. These visual insights help researchers interpret the data before conducting further statistical analysis.

All four types of EDA—univariate, bivariate, multivariate, and graphical—allow researchers to summarize, explore, and interpret data effectively, ensuring it is clean, meaningful, and ready for deeper analysis to generate insights and guide decision-making.

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